AirClay: Few-Shot, Depth-Aware Gesture Control for 3D Sculpting in Blender

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Abstract

We propose AirClay, a markerless two-hand interface for sculpting a deformable mesh in Blender with a commodity RGB webcam. The system will solve two coupled vision problems: causal recognition of four continuous interaction intents (*idle*, *sculpt*, *orbit*, and *resize*) and estimation of the hand's short-term displacement in camera space. Public gesture and 3D hand-pose datasets will provide most of the training data, while one, five, or ten demonstrations per AirClay command will personalize a prototype-based temporal classifier. We will compare rules, a single-frame MLP, a causal GRU trained from scratch, public-data pretraining, and few-shot adaptation. We will separately compare geometric, hand-specific learned, and generic monocular depth cues. Evaluation will measure recognition quality, false activations, 3D displacement error, latency, and performance on a fixed Blender sculpting task. Our main question is whether public-data pretraining and few-shot personalization can enable reliable, depth-aware two-hand sculpting while preserving interactive response time.

1 Problem Statement

Three-dimensional content is increasingly central to animation and games, industrial and product design, digital fabrication, scientific visualization, and immersive AR/VR experiences. Yet producing that content remains a skilled task. In conventional desktop software, a user controls a 3D surface through a 2D mouse or tablet while repeatedly coordinating camera navigation, brush parameters, keyboard modifiers, and editing modes. These tools provide high precision, but their indirect mapping can make rapid shape exploration less natural: the creator's intended hand motion must first be translated into a sequence of device movements and mode changes.

Bare-hand interaction offers a complementary way to specify shape. It can map the motion of the fingers and palms directly to the location, direction, and magnitude of an edit; it also permits asymmetric bimanual interaction, where one hand edits while the other controls the view or scale. Using an ordinary RGB webcam would avoid instrumented gloves, controllers, and dedicated depth or motion-capture hardware, lowering the equipment and setup requirements for casual modeling and classroom use. These are potential interaction benefits, however, and they depend on sufficiently reliable perception. A small number of visually intuitive commands is only useful if natural hand motion does not silently alter the model.

Existing approaches leave several gaps. Gesture-controlled deformation systems often simplify perception with colored gloves or other markers (Mahdikhanlou and Ebrahimnezhad, 2022). Clay-motion recognition has been studied with accurate multi-camera Vicon trajectories (Veldhuijzen et

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al., 2024), but such inputs do not represent the noise, occlusion, and missing detections of a consumer webcam. Many gesture benchmarks classify pre-segmented clips or a fixed closed set, whereas an editor receives an unsegmented stream containing transitions and unknown actions. Their clip-level accuracy therefore does not measure the false activations, unintended releases, or onset delay that directly affect an editing session. Finally, monocular hand-pose methods can recover articulation, but absolute scale and camera-space depth remain ambiguous from a single RGB view, especially when two hands overlap or move toward the camera (Huang et al., 2023; Yu et al., 2023).

AirClay will address these gaps as a causal, depth-aware control problem. Given RGB frames observed up to time t, the system will estimate an intent $y_t \in \{\textit{idle}, \textit{sculpt}, \textit{orbit}, \textit{resize}\}$, a short-term camera-space displacement $\Delta\mathbf{p}_t = (\Delta x_t, \Delta y_t, \Delta z_t)$, and a confidence value. The dominant hand will pinch and drag to deform a local surface region, the non-dominant hand will orbit the model, and a two-hand pinch-and-spread motion will resize the brush. Low-confidence predictions and sustained hand loss will pause or roll back an operation instead of modifying the mesh.

We will test two linked hypotheses. First, a causal temporal encoder pretrained on public gesture data and adapted with one, five, or ten user demonstrations will reduce false activations and premature releases relative to thresholds, a single-frame MLP, and a GRU trained only on the small target set. Second, a calibrated hand-specific depth estimator will recover forward/backward motion more accurately and with less jitter than planar control, inverse palm scale, or a generic monocular depth model. The downstream test will require a user to create two protrusions, rotate the object, resize the brush, undo one edit, and save the result in Blender. Thus the project will connect recognition and depth errors to task completion time, edit accuracy, undo count, and interaction latency rather than evaluating perception only in isolation.

1.1 Related Work

Continuous and dynamic gesture recognition. Jester contains 148,092 short RGB clips of 27 gestures collected in front of laptop or webcam cameras, including explicit non-gesture classes (Materzynska et al., 2019). IPN Hand instead targets continuous recognition: its 50 subjects perform 4,218 gestures in more than 800,000 frames with natural motion between commands (Benitez-Garcia et al., 2021). The recent IPN HandS extension adds revised gesture categories, refined temporal boundaries, and frame-level 21-landmark hand annotations produced through MediaPipe initialization and manual refinement (Benitez-Garcia et al., 2025). These datasets make it possible to learn temporal features and rejection behavior before collecting application-specific data. HaGRID provides complementary diversity in hand appearance, background, illumination, and camera distance across tens of thousands of people (Kapitanov et al., 2024). Recent multimodal attention work further shows the value of combining skeleton and appearance features for dynamic gestures (Balaji and Prusty, 2024). However, most benchmarks report clip accuracy; AirClay will also evaluate false activations, release errors, and latency in an unsegmented interactive stream.

Clay motion and direct 3D manipulation. The SHREC 2024 clay-motion benchmark is the closest domain-specific work. It contains 62 motions in seven fine-grained pottery classes, recorded from one expert potter using 14 Vicon cameras and 28 markers across both hands (Veldhuijzen et al., 2024). Its methods include recurrent networks, HMMs, spatiotemporal graph networks, and a skeleton masked autoencoder. This study demonstrates that clay motions are long, precise, and visually similar, but it classifies isolated motion-capture sequences and does not control an editor. Mahdikhanlou and Ebrahimnezhad deform a 3D object with dynamic gestures from an RGB camera, but use colored gloves to simplify hand observation (Mahdikhanlou and Ebrahimnezhad, 2022). AirClay will instead use bare hands, continuous causal recognition, and real Blender mesh operations.

Few-shot and user-defined gestures. Few-shot learning is attractive because a fixed public taxonomy cannot encode every editor command or a user's preferred motion. Liu et al. combine MediaPipe landmarks, lightweight autoencoders, attention, and similarity-based classification for few-shot continuous RGB gestures (Liu et al., 2022). Schluesener and Buecker train LSTM Relation Networks episodically on landmark sequences derived from Jester, then recognize new gesture sets from one, two, or five examples (Schluesener and Buecker, 2022). Zhou et al. address open-set hand gestures with viewpoint-independent features and cosine-based few-shot incremental classification (Zhou et al., 2025). Building on these ideas, AirClay will learn a general motion embedding from public datasets and adapt only a small projection head and class prototypes from user demonstrations.

Monocular depth and interacting hands. MediaPipe provides useful hand-centered 3D landmarks, but each hand's local coordinates do not determine a common camera-space translation. Absolute depth from one RGB view is inherently ambiguous. MobRecon demonstrates fast single-view hand-mesh recovery with temporal coherence (Chen et al., 2022), while Neural Voting Field directly estimates camera-space 3D joints and global hand geometry (Huang et al., 2023). For two hands, ACR explicitly handles mutual occlusion, truncation, and cross-hand confusion (Yu et al., 2023). Generic monocular models such as Depth Anything V2 offer robust relative scene depth but are not designed for millimeter-scale hand motion (Yang et al., 2024). We will therefore compare a generic depth model with a geometric palm-scale baseline and a compact hand-specific depth regressor, and will focus on calibrated short-term displacement rather than claiming uncalibrated metric depth.

2 Data

Public data will supply broad appearance, motion, and depth supervision. A small AirClay adaptation set will be used only for editor-specific semantics and per-user scale calibration. RHD supplies paired RGB, camera calibration, depth, and 3D joints (Zimmermann and Brox, 2017), while InterHand2.6M supplies camera-space annotations for interacting hands (Moon et al., 2020). Table 1 summarizes the planned sources.

Dataset	Size and annotation	Planned use
Jester	148,092 RGB clips, 27 classes, 22.8 GB	Pretrain dynamic motion and gesture rejection from webcam-like video.
IPN Hand	50 subjects; 4,218 gesture instances; >800K RGB frames; temporal gesture annotations	Pretrain continuous recognition and rejection using landmarks extracted with a pinned MediaPipe version.
IPN HandS	14 gesture classes plus non-gesture; 7,770 total segments including non-gesture; refined boundaries and 21-landmark hand annotations	Optional annotation enhancement, subject to obtaining and validating the corresponding release.
SHREC 2024 Clay	62 sequences, 7 pottery classes, bilateral 28-marker 3D tracks	Domain-specific contrastive pretraining on subtle two-hand motion.
RHD	41,258 train and 2,728 test samples; RGB, depth, intrinsics, segmentation, and 3D joints; 7.1 GB	Train and test an RGB/landmark-to-camera-depth regressor.
InterHand2.6M	2.59M single- and interacting-hand RGB frames with camera and 3D joint annotations	Use a fixed sample from the official 5 FPS interacting-hand subset for two-hand depth and occlusion tests.
HaGRID	Large RGB gesture collection across subjects, scenes, and 0.5–4 m distance	Use public weights or a fixed class-balanced subset for appearance robustness.

Table 1: Public datasets and their distinct roles. Large optional datasets will be sampled with stored frame IDs so that experiments remain reproducible.

The core reproducible pipeline will use Jester, the original IPN Hand RGB videos and temporal annotations, SHREC 2024, and RHD. IPN HandS will be an optional enhancement after we obtain its annotations and verify their alignment with the corresponding videos. We will record the annotation version, file list, and preprocessing configuration for reproducibility. We will not require the full 1.5 TB HaGRIDv2 release; we will use public weights or selected classes. Likewise, InterHand2.6M will be restricted to a fixed, subject- and view-stratified sample of its 5 FPS interacting-hand data. Every RGB dataset will be processed by one pinned MediaPipe version (Zhang et al., 2020) into a common representation with up to 21 image landmarks per hand, timestamps, handedness, and validity masks.

When dataset-provided skeleton annotations are available, we will retain them separately from our MediaPipe predictions. IPN HandS annotations are initialized with MediaPipe and manually refined using editing and interpolation tools; we will use them as curated landmark annotations, not as independently measured metric 3D ground truth. Dataset-native 3D coordinates will remain in separate fields with their camera calibration; incompatible coordinate systems will not be concatenated as if they shared a scale.

Public labels will supervise reusable primitives—idle, pinch/click, translation, push/pull, and two-hand scale—rather than being relabeled directly as Blender commands. For each target user, we will

then record one, five, or ten demonstrations of each AirClay intent plus two minutes of natural hand motion. Three short calibration sequences at approximately 30, 45, and 60 cm will estimate person- and camera-specific scale. Support, validation, and query sequences will be separated before sliding windows are generated, and the final test will hold out people rather than frames.

3 Method

3.1 Representation and Public-Data Pretraining

At roughly 30 Hz, the input will include wrist-centered, palm-scale-normalized landmarks; wrist and fingertip image positions and velocities; pinch ratios; two-hand distance and relative velocity; hand-presence masks; and lightweight features from hand RGB crops. All online features will use only the current and previous frames. A causal GRU or temporal convolutional network (TCN) will encode a 24-frame window into a motion embedding. We will first train this encoder on the Jester and IPN primitives with a supervised classification loss. SHREC sequences will add a contrastive objective so that clips from the same fine-grained clay class remain close despite duration changes. Public non-gesture segments and failed detections will supply rejection examples.

3.2 Few-Shot Intent Recognition

Our primary few-shot model will use a Prototypical Network. If $f_\theta(x)$ is the temporal embedding for sequence x, the prototype for class c will be

$$\mathbf{p}_c = \frac{1}{|S_c|} \sum_{x_i \in S_c} f_\theta(x_i), \tag{1}$$

where S_c is the user's support set. Prediction will use cosine distance to the four prototypes. A validation-set distance threshold will reject low-confidence windows as *idle*. During personalization, most of the public-data encoder will remain frozen; only the projection layer and class prototypes will be updated. Episodic training on public gesture classes will make the training objective match 1-, 5-, and 10-shot evaluation.

We will compare five recognition approaches: pinch thresholds and hand-role rules; a single-frame MLP; a causal GRU trained from scratch on the same support data; the public-pretrained GRU with a linear classifier; and the pretrained GRU with prototype adaptation. All approaches will feed the same causal state machine. The state machine will use probability smoothing, onset/offset hysteresis, a short two-hand arbitration window, mode locking, and rollback after sustained hand loss. This isolates the recognition model as the experimental variable.

3.3 Depth and Continuous Motion

Intent classification will be separated from continuous control. The latter will estimate short-term camera-space displacement

$$\Delta\mathbf{p}_t = (\Delta x_t, \Delta y_t, \Delta z_t). \tag{2}$$

Image-plane landmark motion and the calibrated camera projection will provide Δx_t and Δy_t. We will compare three estimates of Δz_t: (1) inverse apparent palm width after three-distance calibration; (2) a compact regressor trained on RHD and sampled InterHand RGB, 2D joints, and camera-space root depth; and (3) relative depth sampled from Depth Anything V2 Small inside the hand region. The learned regressor will predict both depth and confidence. Low-confidence frames will retain planar motion but suppress depth updates. Temporal filtering and maximum-velocity limits will prevent a single unstable estimate from causing a large mesh edit.

Blender will receive intent, normalized pointer position, displacement, confidence, sequence number, and operation ID over local UDP. In sculpt mode, a view ray will select the surface, and the camera-space displacement will be mapped to the current view basis. Nearby vertices will move with a smooth geodesic falloff from a snapshot taken at gesture onset. Orbit mode will map motion to camera azimuth and elevation, while resize mode will map the change in bilateral 3D distance to brush radius. Completed operations will enter a 30-step undo/redo history.

3.4 Evaluation

For recognition, we will report macro-F1, per-class precision and recall, unknown-motion rejection, false activations per minute, missed operations, premature releases, mode-switch errors, and onset/offset latency. Few-shot results will be reported at 1, 5, and 10 shots on held-out users. For depth, we will report wrist/root depth mean absolute error, 3D joint MPJPE where ground truth is available, forward/backward direction accuracy, temporal jitter, and inference time.

Finally, participants will perform a fixed Blender task: create two protrusions on a sphere, rotate the model, resize the brush, correct one edit with undo, and save the result. We will compare planar-only, geometric-depth, and learned-depth control using task completion time, success rate, undo count, overshoot count, and final control-point error. The target is at least 20 Hz interaction updates and median end-to-end latency below 250 ms; these are performance goals to be measured, not assumed results.

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